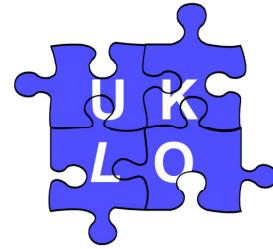


UKLO Round 2 March 2015

Data and questions

[Please use a **black** pen so that we can scan your answer.]



Maxakalí

Maxakalí is spoken in the eastern Brazilian region of Minas Gerais by about 1200 people. In this problem your task is to match words and phrases to their English meanings, whose identifying letter (A, B, etc) you should write in the first column before the corresponding Maxakalí expression.

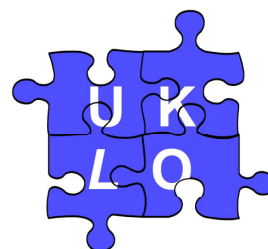
Maxakalí		English	
	ka'õgãhã	A.	cover
	ka'ok	B.	eye
	kuxa	C.	eyelid
	kuxa ka'ok	D.	foot
	mĩkax	E.	go
	mĩkaxxax	F.	hard
	mĩptut	G.	harden
	mĩptut mōg	H.	heart
	mĩptut mōg kuxa	I.	house
	mĩptut mōg pata	J.	knife
	mōg	K.	knife sheath
	mōgãhã	L.	lead
	pa	M.	motor
	pa ka'ok	N.	motor vehicle
	pata	O.	shoe
	pataxax	P.	stubborn
	paxax	Q.	tire
	xax	R.	wide awake

Pronunciation Notes: A tilde (~) above a vowel means that it is pronounced nasally. The letter <x> represents a sound like English 'sh,' and the apostrophe ' represents a glottal stop. Other letters are reasonably similar to their equivalents in English.

UKLO Round 2 March 2015

Malagasy Crossnumber

The main task is to solve a crossnumber puzzle in Malagasy, an Austronesian language and one of the official languages of Madagascar, but we must first explain what a crossnumber puzzle is, using an example from English. This type of puzzle is much like a crossword, except that each square in the grid is meant to hold a single digit rather than a single letter, and the clues are trivially easy provided you know the language. For example, if the clue for 'one across' is 'Seventy-four', you just write '7' in cell 1 and '4' in the next square to the right; and if 'one down' is 'seventy-three', you add '3' in the cell below.



But what if you don't know the language? Here is a crossnumber puzzle with the clues written in Malagasy. You need to know that:

- the clues are not given in the same order as the answers in the grid.
- no answer has zero as its first digit.

Clues: ACROSS

ANSWER #	ANSWER
Dimy	Efatra amby enimpolo sy telonjato sy sivy alina
Fito	Dimy ambin'ny folo sy roanjato sy arivo sy fito alina
Folo	Iraika amby fitopolo
Iray	Fito amby fitopolo
Sivy	Folo
Telo	Fito ambin'ny folo
Valo	Valo amby enimpolo sy sivinjato sy dimy arivo sy alina

Clues: DOWN

ANSWER #	ANSWER
Efatra	Iraika amby valopolo sy dimanjato sy efatra arivo sy fito alina
Enina	Sivy amby roapolo sy telonjato
Iray	Iraika ambin'ny folo sy fitonjato sy sivy arivo sy fito alina
Roa	Dimampolo sy zato sy fito alina
Telo	Fito amby enimpolo sy zato sy enina arivo sy alina

1	2		3	4
5		6		
7				
8				
9			10	

Q.2.1. Solve this crossnumber problem by filling all the cells.

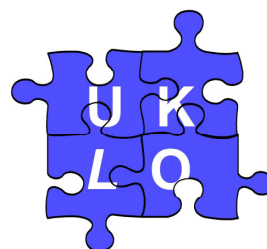
Q.2.2: Write the following numbers in Malagasy.

digits	Malagasy words
7	
15,968	
99,573	
80,638	
817	

UKLO Round 2 March 2015

1. Do-This-Do-That in Hmong

Hmong is a language of Southern China and Southeast Asia, especially the countries of Vietnam, Laos, and Thailand. One dialect of Hmong that is spoken in all of these countries is called Hmong Daw (or White Hmong). The sentences and phrases that you see below are from Hmong Daw, written in a writing system called RPA (Romanized Popular Alphabet). In this writing system, a doubled vowel character is pronounced as nasalized or with a following “ng” sound, and a following consonant symbol is used to show the tone of the syllable (high, mid, low, rising, or falling pitch). Hmong Daw has seven tones, one of which is indicated by the absence of any symbol. For instance, in this system, Hmong is written as *Hmoob*.



The table on the next page contains 31 Hmong Daw sentences and phrases, and 31 English translations. Note that:

- the Hmong expressions and their English translations are in a different order.
- within a Hmong expression and its translation, words may be arranged differently.
- an English word may translate more than one word in Hmong, and vice versa.
- the abbreviations *sg*, *du* and *pl* mean ‘singular’, ‘dual’ (two) and ‘plural’ (more than two).

Q.3.1. Pair each Hmong expression with its English translation by writing the latter’s number in the first column of the table.

Q.3.2. Translate the following into Hmong:

Lao and Hmong people	
behave as primates	
have precious metals	

#	Hmong	#	English
	Neeg them nyiaj rau koj.	1.	a very tall tree
	Nej sib pom lawm.	2.	behave as one group
	phem dab phem tuag	3.	grass sprouts
	Kuv lub tsev nqeeb phem heev.	4.	He planted a grass plant.
	Ua ib pawg.	5.	People fear spirits.
	Wb sib ntsib.	6.	some legal proceedings
	Khib kuv heev.	7.	He is very upset.
	Dab tsis ntshai.	8.	It offends me very much.
	Nwg cog ib tsob tauj.	9.	We (du.) are very ugly.
	Nej tsis tau pom kuv.	10.	speak Hmong language
	Ntuj no lawm.	11.	Heaven is cold now.
	Neeg Nplog pe mlom.	12.	petition deities
	tej plaub tej ntug	13.	a band of monkeys
	Kuv tsis tau ntsib nwg.	14.	You (pl.) did not see me.
	tsev hais plaub	15.	You (pl.) see each other now.
	neeg loj neeg siab	16.	ugly as spirits and death
	hais lus Hmoob	17.	People pay silver to you.
	sib ntsib sib pom	18.	Lao people reverence images.
	tsis khib tsis chim	19.	not angry or offended
	pe dab pe mlom	20.	My grass house is very ugly.
	ib pab liab	21.	big and tall people
	Koj muaj kub.	22.	Apes are like people.
	ntau pab ntau pawg	23.	Spirits dont fear.
	Nej muaj tej tsev loj.	24.	You (sg.) have gold.
	Wb phem heev.	25.	many factions
	Neeg ntshai dab.	26.	I did not meet him.
	Nwg chim heev.	27.	worship images and spirits
	ntsis tauj ntsis nqeeb	28.	court (house of speaking legal proceedings)
	Cuam zoo li neeg.	29.	encounter one another
	thov ntuj thov dab	30.	You (pl.) have some big houses.
	ib tsob ntoo siab heev	31.	We (du.) met each other.



Aymara rules

This problem concerns Aymara, an indigenous language spoken in Bolivia, but it also raises general questions about the kinds of rules that are possible for ‘phonotactics’ – for explaining how sounds should be put together to make words. Not just any combination of sounds can be a word; for instance, in English *blid* is a possible word (although it doesn’t actually exist), but *bnid* is impossible because it breaks the phonotactic rules of English. There are many types of rules one can write to describe sound patterns. In this problem we consider three rule types:

Type A: These constraints require that a certain sound occur either an even or odd number of times. We write Type A constraints as follows:

either **X=Even**, meaning the sound X must occur an even number of times,
or **X=Odd**, meaning the sound X must occur an odd number of times.

For example, b=Even requires that there be an even number of b’s in every word. Thus, b=Even rules out words like “bas” or “bisbanib” while allowing words like “tas” or “bistanib”. The rule b=Odd would do the exact opposite: allow “bas” and “bisbanib” but rule out “tas” and “bistanib”.

Type B: These constraints prohibit a certain sound from occurring right before another sound. We write Type B constraints as follows:

***XY**, meaning that X must not occur right before Y.

For example, *bn prohibits b from immediately preceding n. Thus, *bn rules out “abnik” as a possible word while allowing words like “atnik” or “anbik”.

Type C: These constraints prohibit a certain sound from occurring before another sound at any point in the word. We write Type C constraints as follows:

***[XY]**, meaning that X must not occur before Y, no matter what may or may not come in between.

For example, *[bn] prohibits b from preceding n anywhere. Like *bn, *[bn] allows words like “atnik” or “anbik” and rules out “abnik”, but unlike *bn, *[bn] also rules out “abitnik”.

In this problem, you will write constraints in the forms just described in order to account for some of the phonotactics of Bolivian Aymara. The following tables list acceptable and unacceptable words in Aymara:

Acceptable

kawki	sipita	pisi	janana	purapa	alwa	qawa
kikpa	oqara	kunka	tarkaka	seqe	qemi	putu
qolqeni	yuru	tage	jiliri	qapa	jaqet	mayni
toqe	nayra	tukjata	tiwula			

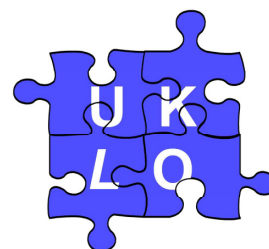
Unacceptable

ekawu	okatu	qaqira	kutaqa	qulwaqa	maqetug	toqaki
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Q.4.1. write a set of constraints that will rule out all of the bad forms while permitting all of the good forms. Use as few constraints as possible. Follow the format described above for writing constraints. Note: you don’t need to account for every pattern you might find. You will lose points only for ruling out the acceptable forms, failing to rule out the unacceptable forms, or using more constraints than necessary.

Q.4.2. Out of these three types of rules, only two are known to be needed when describing the phonotactics of human languages. Which type of constraint isn't needed for the Aymara data?

Data and questions



Vulgar Latin and the Romance languages

From 100-300 CE the Roman Empire included much of Europe, and the use of Latin spread through this empire, mainly in a spoken version called “Vulgar [= colloquial] Latin”. Later, as the empire declined, different dialects of Vulgar Latin changed and diverged so much that they became different languages: the “Romance” languages of today’s Europe, including Italian, Spanish, Catalan (spoken in the north-east of Spain), French and Romanian. English is distantly related to Latin, so there are many basic words in the two languages that have a vague resemblance; and later (in the Middle Ages) English borrowed many more words directly from Latin and from the Romance languages.

The following tables show pronunciations rather than spellings, so each letter indicates one sound only (e.g. g is always pronounced as in ‘get’, and never as in ‘gem’); some special symbols are used to show particular sounds:

- u = the sound of “oo” in “boot”.
- ə = the sound of the “a” in “about”.
- ɪ = a sound between ə and i.
- j = the sound of “y” in “yes”.
- ʌ = like the “lli” sound in “million”.
- ʃ = the sound of “sh” in “ship”.
- dʒ = the first sound of “jump”.
- œ = similar to ə but with the lips rounded like an “o”.
- ŋ = the sound of “ng” in “bang” (where some speakers of English also pronounce a distinct g).
- tʃ = the sound of “ch” in “chip”.
- y = like the sound of “ee” in “bee”, but with the lips rounded .
- ɥ = a quick version of y.
- ð = the sound of “th” in “other”.
- The symbol ~ above a vowel means that the vowel is nasalized (sounds through the nose).
- For words of more than one syllable, the symbol ' means that the stress is on the following syllable.

Note that changes in the Romance pronunciations are not always predictable; and also that occasionally a Romance language has used a different Latin word, or even one from another language clearly not related to Latin.

Table 1 shows the pronunciation of some corresponding words in Vulgar Latin and in three modern Romance languages.

Table 1

Table 1 Vulgar Latin	A	B	C	English translation
'matrem	mer	'marə	'mamə	mother
'patrem	per	'parə	'tatə	father
'filium	fis	fiλ	'fiul	son
'filiam	fij	'fiλə	'fiika	daughter
so'rorem (or ger'manam)	sœr	dʒər'manə	'sora	sister
'fratrem (or ger'manum)	frer	dʒər'ma	'frate	brother
'katum	ʃa	gat	pi'sika	cat
'kanem	ʃjē	ka	'kijne	dog

Q.5.1. Put the words listed below into their correct places in table 2.

bin, 'dekem, dew, dœ, dos, 'dowə, katr, 'kentum, 'kwatrə, 'kwatuor, næf, 'novem, 'nowə, 'okto, opt, sā, sēk, seks, sen, 'septem, set, siŋ, sis, sis, 'ʃapte, tres, tres, 'tʃintʃi, u, 'unum, ʒit, vē, 'zetʃe

Table 2

Vulgar Latin	A	B	C
	œ		'unul
'dues			
	trwa		trej
			'patru
'kwinkwe			
			'ʃase
		set	
		bwit	
		now	
	dis		
'viginti			dowə'zetʃi
			'sutə

Q.5.2. Put the words below into their correct places in table 3:

Romance words

'apə, beλ, ə'skriwrə, flœr, 'floare, 'hominem, kənt'a, 'luna, 'lunə, o, om, 'skribere, ʃă'te, 'veki, vi, 'viatsə

English words

flower, life, man, moon, sing, water, write

Table 3

Vulgar Latin	A	B	C	English translation
'akwa		'ajgwə		
kan'tare			kin'ta	
'flore		flo		
	e'krir		'skrie	
		'omə	om	
'vita		'biðə		
	lyn		'lunə	
'veklus	vjœ			old (masculine)

Q.5.3. Write the missing pronunciations or meanings in table 4:

Table 4

Vulgar Latin	A	B	C	English translation
'panem	pē			bread
	e'twal	ə'streʎə	'stele	star
'vekla	vjej		'veke	
	œj	uʎ	'oki	eye
'videt	vwa	bew		sees
'videre	vwar		ve'dea	
	kã	kwan	kɪnd	when?

Q.5.4. Explain as clearly and concisely as you can how pronunciations changed between Vulgar Latin and the three languages.